Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 5

Data Management Policy

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Learning Objectives

1. Differentiate and adhere to requirements embodied in funder, institutional and publisher policies
2. Explain ownership considerations related to data sharing
3. Describe potential legal issues connected to one's data (intellectual property, copyright, licensing, data retention)

RESEARCHER DATA SHARING INSIGHTS

WILEY

- Wiley's Researcher Data Insights Survey was launched earlier this year to understand how and why researchers make their research data publicly available. The study's results, highlighted below, are intended to advance the global conversation about data sharing and help Wiley better meet the needs of our researchers, authors, and partners in the rapidly evolving landscape of scientific research and communications.
- The survey was deployed in March 2014 and received more than 2,250 responses from researchers around the world.

GLOBAL DATA SHARING TRENDS

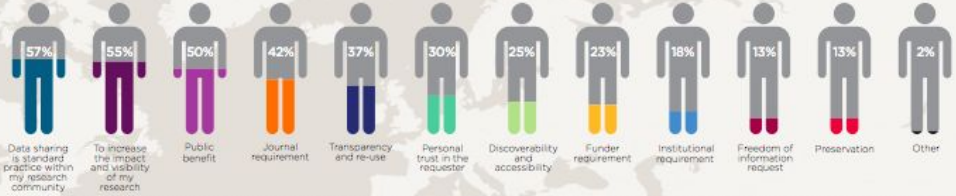
Data sharing practices vary widely across research fields and geographic areas. Just over half of researchers report making their data publicly available, though archiving results in repositories is not yet the norm.

WAYS DATA IS SHARED

- **67%** As supplementary material in a journal
- **37%** Personal, institutional or project webpage
- **26%** Institutional data repository *(i.e. university or institute-sponsored)*
- **19%** Discipline-specific data repository
- **6%** General-purpose data repository *(e.g. Dryad, figshare)*
- ✓ **5%** Other

Globally, researchers also report sharing their data in limited and non-permanent ways: 57% are sharing data at a conference while 42% of researchers share their data upon informal request (e.g. email, direct contact, etc.).

RESEARCHER MOTIVATIONS FOR SHARING DATA



DATA SHARING TRENDS BY COUNTRY

46% SHARING / 54% NOT SHARING

UNITED STATES
Among researchers in the US sharing their data publicly, two out of three do so because it is standard practice in the communities and because they believe it benefits the public. Similar to their counterparts in the UK, the majority of US-based researchers also share data to increase the impact or visibility of their research.

43% SHARING / 57% NOT SHARING

UNITED KINGDOM
While more than 40% of UK researchers are sharing data, only about 14% are using discipline-specific or other public repositories like Dryad and figshare. The two key drivers that motivate UK researchers to share their data are the prospect of gaining increased impact or visibility for their work and to satisfy funder requirements.

44% SHARING / 56% NOT SHARING

JAPAN
Compared with their counterparts around the world, researchers in Japan cite concerns about being scooped as a reason for not sharing data more frequently. Nearly five out of ten Japanese researchers point to this as a reason for not sharing their data, roughly double the global average.

36% SHARING / 64% NOT SHARING

CHINA
Nearly five in ten Chinese researchers say they are not sharing data because they are not required to do so by their funders or institutions. They are more likely than their global counterparts to say that they do not see data sharing as a personal responsibility and plan to take direction from funders to guide their data sharing decisions in the future.

52% SHARING / 48% NOT SHARING

BRAZIL
Two out of three researchers in Brazil say that a guarantee of proper credit or attribution would compel them to share more of their data publicly in the future.

41% SHARING / 59% NOT SHARING

AUSTRALIA
Researchers in Australia say they would be most incentivized to make their data accessible in the future to ensure preservation as well as transparency and re-use. The majority of researchers also ranked funder requirements among top reasons to share in the future.

55% SHARING / 45% NOT SHARING

GERMANY
Among German researchers sharing their data publicly, three out of four are driven to share data because they believe it will increase the visibility of their research and want to ensure public transparency and re-use. About 20% of German researchers are making use of general purpose repositories (like figshare and Dryad), significantly more than their counterparts around the world, including those in the US and UK.

REASONS WHY RESEARCHERS ARE HESITANT TO SHARE THEIR DATA

- **42%** Intellectual property or confidentiality issues
- **36%** My funder/institution does not require data sharing
- **26%** I am concerned that my research will be scooped
- **26%** I am concerned about misinterpretation or misuse
- **23%** Ethical concerns
- **22%** I am concerned about being given proper citation credit or attribution
- **21%** I did not know where to share my data
- **20%** Insufficient time and/or resources
- **16%** I did not know how to share my data
- **12%** I don't think it is my responsibility
- **12%** I did not consider the data to be relevant
- **11%** Lack of funding
- **7%** Other

DATA SHARING BY DISCIPLINE

Data sharing, specifically by way of data repositories, is most prevalent amongst life scientists, particularly those in the earth and environmental and agriculture and food sciences.

Where Health Scientists share their work:

- **68%** As supplementary material in a journal
- **29%** Personal/institutional/lab webpages
- **29%** Institutional data repositories *(i.e. university or institute-sponsored)*
- **21%** Discipline-specific data repositories
- **5%** General-purpose data repositories *(e.g. Dryad, figshare)*

A typical *Health Science researcher* says she would be motivated to share her data in the future in order to benefit the public, so long as privacy and ethical concerns are assuaged.

Where Life Scientists share their work:

- **76%** As supplementary material in a journal
- **42%** Discipline-specific data repositories
- **29%** Personal/institutional/lab webpages
- **23%** Institutional data repositories *(i.e. university or institute-sponsored)*
- **13%** General-purpose data repositories *(e.g. Dryad, figshare)*

A typical *Life Science researcher* says she would be motivated to share more of her data in the future if she was guaranteed proper credit.

Where Physical Scientists share their work:

- **69%** As supplementary material in a journal
- **41%** Personal/institutional/lab webpages
- **28%** Institutional data repositories *(i.e. university or institute-sponsored)*
- **10%** Discipline-specific data repositories
- **3%** General-purpose data repositories *(e.g. Dryad, figshare)*

A typical *Physical Science researcher* says she would be motivated to share her data in the future because it is standard practice within her research community and because it increases the impact and visibility of her work.

Where Social Scientists share their work:

- **52%** As supplementary material in a journal
- **51%** Personal/institutional/lab webpages
- **25%** Institutional data repositories *(i.e. university or institute-sponsored)*
- **3%** General-purpose data repositories *(e.g. Dryad, figshare)*
- **2%** Discipline-specific data repositories

A typical *Social Science and Humanities researcher* says she would be motivated to share her data in the future if it increased the impact and visibility of her work or if she was required to by her funder.

Global Overview of RDM

- Kathleen Shearer “Comprehensive Brief on Research Data Management Policies”
- State of RDM policy development internationally
- Results of an environmental scan from fall/winter 2014-2015
- Up-to-date and detailed overview of the RDM policy environment
- Challenges related to policy adoption

Overview of OSTP Responses

	Policy Coverage		Policy Stipulations					Support Provided			
Funder	Published Outputs	Data	Time Limits	DMP	Access / Sharing	Long-term curation	Monitoring	Guidance	Article Repository	Data Repository	Costs
AHRQ	●	●	●	●	●	◐	◐	◐	●	◐	◐
ASPR	●	●	●	●	●	●	●	◐	●	◐	◐
CDC	●	●	●	●	●	◐	●	◐	●	◐	◐
DOD	●	●	●	●	●	○	●	○	●	**	◐
DOE	●	●	●	●	●	◐	◐	◐	●	◐	◐
DOT	●		◐						●		
FDA	●	●	●	●	●	◐	●	◐	●	◐	◐
NASA	●	●	●	●	●	◐	●	◐	●	◐	◐
NIH	●	●	●	●	●	◐	◐	◐	●		◐
NIST	◐	●	◐	●	●	●			●	**	
NOAA	●	●	●	●	●	●	●	◐	●	◐	◐
NSF	●	●		●	●	◐	●		●	●	●
SI	●	●	●	**	●	○	○	○	◐	◐	○
USDA	●	◐				◐	○	○	●	**	●
USAID	○	●	○		◐		○	○	○	**	
USGS		●		●	◐						
VA	◐	◐	◐	●	**	◐	●	○	●	**	●

Last updated: December 29, 2015

● Full Coverage ◐ Partial Coverage
○ No Coverage ☐ Not Enough Information
** For more information, see http://bit.ly/FedOASummary

Typical Elements of a Funder Data Policy

Policy requirements	
Data quality and standards	Investigators are required to adhere to international standards to enable access and reuse in the discipline. Data documentation and metadata must accompany data so that the data is understandable by others.
Data access and sharing	Investigators are required to make data available to be shared (usually upon publication of results or shortly thereafter, although some agencies do allow embargo periods). Requirements for deposit of metadata into a local or national catalogue
Data retention and preservation	Data should be retained for a certain time limit, where possible, investigators must deposit their data in a long-term archive to ensure the preservation of their data.
Data management plans	Research proposals must include a Data Management Plan in proposal.

Example Institutional Data Policy

Policy

All institutional data is owned by New York University and, as such, all members of the University community are responsible for appropriately respecting and protecting the asset. (Also refer to the *Policy on Responsible Use of NYU Computers and Data*, www.nyu.edu/about/policies-guidelines-compliance/policies-and-guidelines/responsible-use-of-nyu-computers-and-data-policy-on.html.) This policy establishes uniform data management standards and identifies the shared responsibilities for assuring the integrity of the data and that it efficiently and effectively serves the needs of the University. This policy applies to data used in the administration of the University and all of its units, such as schools and administrative and auxiliary units, as well as data reposited in the University Data Warehouse.

New York University

Policy on Retention of and Access to Research Data

Summary: Establishes University policy to assure that Research Data are appropriately recorded, archived for the required time, and available for review under the appropriate circumstances.

Effective Date: March 1, 2010

Issuing Authority: University Provost

Responsible Officer: Senior Vice Provost for Research

Who Needs to Know this Policy: This policy shall apply to all University faculty, staff, postdoctoral appointees, students and any other persons involved in the conduct of research at or under the auspices of the University, and it shall apply to all research projects on which those individuals work, regardless of the source of funding for the project.

Related Policies: Statement of Policy on Intellectual Property; Policy on Tangible Research Property.

Policy Purpose: Maintaining accurate and appropriate research records is an essential component of any research project. It is necessary to support and substantiate findings, to protect intellectual property rights, to facilitate management of the research program of New York University ("NYU" or the "University"), and to ensure compliance with federal regulations and other University policies.

https://www.nyu.edu/about/policies-guidelines-compliance/policies-and-guidelines/retention-of-and-access-to-research-data.html

Publisher Policies

SHERPA/RoMEO

. . . opening access to research

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Publisher:	Elsevier, United States
Journals:	(~2030 journals) (including journals with special policies) - involving 855 other organisations
RoMEO:	This is a RoMEO **green** publisher
Copyright:	Unleashing the power of academic sharing - Sharing Policy - Sharing and Hosting Policy FAQ - Green open access - Journal Embargo Period List (pdf) - Journal Embargo List for UK Authors - Attaching a User License (pdf) - Funding Body Agreements
Updated:	01-May-2015

These are the publisher's default policies. Individual journals may have special permissions, especially if they involve other organisations or have paid open access options. **Always run a journal title or ISSN search to check.**

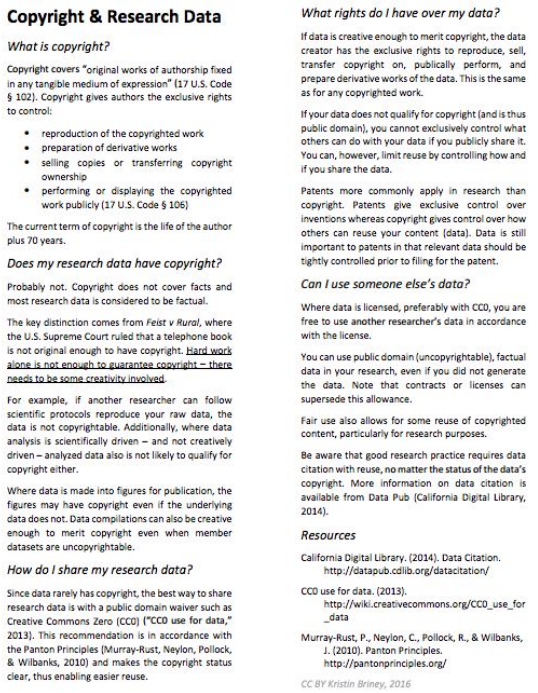
Author's Pre-print:	✓ author **can** archive pre-print (ie pre-refereeing)
Author's Post-print:	✓ author **can** archive post-print (ie final draft post-refereeing)
Publisher's Version/PDF:	✗ author **cannot** archive publisher's version/PDF
General Conditions:	• Authors pre-print on any website, including arXiv and RePEC • Author's post-print on author's personal website immediately • Author's post-print on open access repository after an embargo period of between 12 months and 48 months • Permitted deposit due to Funding Body, Institutional and Governmental policy or mandate, may be required to comply with embargo periods of 12 months to 48 months • Author's post-print may be used to update arXiv and RepEC • Publisher's version/PDF cannot be used • Must link to publisher version with DOI • Author's post-print must be released with a Creative Commons Attribution Non-Commercial No Derivatives License
Mandated OA:	Compliance data is available for 57 funders
Paid Open Access:	Open Access
Notes:	• Publisher last reviewed on 03/06/2015
Exceptions to this policy:	• 12 months - RoMEO Green • Cancer Treatment Communications - RoMEO Green • Creative Commons - RoMEO Green • Creative Commons Attribution - RoMEO Green

Legal Considerations

- Check with your institution's intellectual property office regarding the ownership of research data
- Many policies impact retention of data and related documentation for audits and investigations (IRB, institution, funder, state and federal laws)
- "Copyright does not protect facts, ideas, systems, or methods of operation, although it may protect the way these things are expressed" (U.S. Copyright Office, 2012)

Licensing Data

- Derived data or compiled databases can be licensed
- Creative Commons licenses provide data creators options in which they can choose provisions for their data while enabling dissemination and reuse
- CC0 (Creative Commons Zero) license is often recommended for scientific data
- If you want to use someone else's data — you need to know if it's licensed, and what the terms of the license are



Copyright & Research Data

What is copyright?

Copyright covers "original works of authorship fixed in any tangible medium of expression" (17 U.S. Code § 102). Copyright gives authors the exclusive rights to control:

- reproduction of the copyrighted work
- preparation of derivative works
- selling copies or transferring copyright ownership
- performing or displaying the copyrighted work publicly (17 U.S. Code § 106)

The current term of copyright is the life of the author plus 70 years.

Does my research data have copyright?

Probably not. Copyright does not cover facts and most research data is considered to be factual.

The key distinction comes from *Feist v Rural*, where the U.S. Supreme Court ruled that a telephone book is not original enough to have copyright. Hard work alone is not enough to guarantee copyright – there needs to be some creativity involved.

For example, if another researcher can follow scientific protocols reproduce your raw data, the data is not copyrightable. Additionally, where data analysis is scientifically driven – and not creatively driven – analyzed data also is not likely to qualify for copyright either.

Where data is made into figures for publication, the figures may have copyright even if the underlying data does not. Data compilations can also be creative enough to merit copyright even when member datasets are uncopyrightable.

How do I share my research data?

Since data rarely has copyright, the best way to share research data is with a public domain waiver such as Creative Commons Zero (CC0) ("CC0 use for data," 2013). This recommendation is in accordance with the Panton Principles (Murray-Rust, Neylon, Pollock, & Wilbanks, 2010) and makes the copyright status clear, thus enabling easier reuse.

What rights do I have over my data?

If data is creative enough to merit copyright, the data creator has the exclusive rights to reproduce, sell, transfer copyright on, publically perform, and prepare derivative works of the data. This is the same as for any copyrighted work.

If your data does not qualify for copyright (and is thus public domain), you cannot exclusively control what others can do with your data if you publicly share it. You can, however, limit reuse by controlling how and if you share the data.

Patents more commonly apply in research than copyright. Patents give exclusive control over inventions whereas copyright gives control over how others can reuse your content (data). Data is still important to patents in that relevant data should be tightly controlled prior to filing for the patent.

Can I use someone else's data?

Where data is licensed, preferably with CC0, you are free to use another researcher's data in accordance with the license.

You can use public domain (uncopyrightable), factual data in your research, even if you did not generate the data. Note that contracts or licenses can supersede this allowance.

Fair use also allows for some reuse of copyrighted content, particularly for research purposes.

Be aware that good research practice requires data citation with reuse, no matter the status of the data's copyright. More information on data citation is available from Data Pub (California Digital Library, 2014).

Resources

California Digital Library. (2014). Data Citation. http://datapub.cdlib.org/datacitation/

CC0 use for data. (2013). http://wiki.creativecommons.org/CC0_use_for_data

Murray-Rust, P., Neylon, C., Pollock, R., & Wilbanks, J. (2010). Panton Principles. http://pantonprinciples.org/

CC BY Kristin Briney, 2016

Briney, Kristin. 2016. "Data and Copyright." *Figshare*. http://dx.doi.org/10.6084/m9.figshare.3117838.v1